Last Time:

- Inverse-Square Law
- Newtonian Gravity
- Elliptical Orbits

Today:

- Conserved Quantities
- Eccentricity Vector
- Elliptical Orbits Pt. 2
- Orbital Elements

Conserved Quantities:

- Last time we saw that the (specific) angular momentum vector is conserved:

$$\vec{h} = \vec{r} \times \vec{v}$$

$\vec{h}$: normal to orbit plane; $\vec{r}$: in orbital plane

- Since gravity is a conservative force, the (specific) energy is also conserved:

$$E = T + U = \frac{1}{2}\vec{v}\cdot\vec{v} - \frac{\mu}{r}$$

- $\vec{h}$ specifies orientation of the orbital plane
- Energy tells us the size of the orbit.
- We're still missing a conserved quantity that tells us the shape of the orbit and its orientation within the orbit plane.

Eccentricity Vector:

- We want a conserved vector that lives in the orbit plane.
- Lets look at $\vec{v} \times \vec{h}$:

$$\frac{d}{dt}\left(\vec{v} \times \vec{h}\right) = \dot{\vec{v}} \times \vec{h} = \left(\frac{-\mu}{r^3}\vec{r}\right) \times \vec{h} = \frac{-\mu}{r^3}\left(\vec{r} \times \vec{r} \times \vec{v}\right)$$

- Use vector identity: $\vec{a} \times \vec{b} \times \vec{c} = \vec{b}(\vec{a} \cdot \vec{c}) - \vec{c}(\vec{a} \cdot \vec{b})$

$$\frac{d}{dt}\left(\vec{v} \times \vec{h}\right) = \frac{\mu}{r^3}\left(\vec{v}(\vec{r} \cdot \vec{r}) - \vec{r}(\vec{r} \cdot \vec{v})\right) = \frac{\mu}{r^3}\left(\vec{v}\, r^2 - \vec{r}\, r \dot{r}\right)$$

$$= \mu\left(\frac{\vec{v}}{r} - \frac{\vec{r}\,\dot{r}}{r^2}\right)$$

$\frac{d}{dt}\left(\frac{\vec{r}}{r}\right) = \dot{\hat{r}}$

$$\Rightarrow \frac{d}{dt}\left(\vec{v} \times \vec{h} - \mu\frac{\vec{r}}{r}\right) = 0$$

This is conserved - "Laplace-Runge-Lenz" vector

- Define eccentricity vector (unitless):

$$\vec{e} = \frac{\vec{v} \times \vec{h}}{\mu} - \frac{\vec{r}}{r}$$

$\vec{r} = \hat{r}$

- $\vec{e}$ points from the origin / central body to periapsis (perigee for Earth), and $\|\vec{e}\|$ = eccentricity of ellipse.

Elliptical Orbits Pt. 2

- If we calculate $\vec{r} \cdot \vec{e}$

$$\vec{r} \cdot \vec{e} = \frac{\vec{r} \cdot (\vec{v} \times \vec{h})}{\mu} - \frac{\vec{r} \cdot \vec{r}}{r}$$

- Can cyclically permute vectors: $\vec{a} \cdot (\vec{b} \times \vec{c}) = \vec{c} \cdot (\vec{a} \times \vec{b}) = \vec{b} \cdot (\vec{c} \times \vec{a})$

$$\Rightarrow \vec{r} \cdot \vec{e} = \frac{\vec{h} \cdot (\vec{r} \times \vec{v})}{\mu} - r = \frac{h^2}{\mu} - r$$

$$\Rightarrow r = \frac{h^2}{\mu} - \vec{r} \cdot \vec{e}$$

$$\Rightarrow r^2 = \vec{r} \cdot \vec{r} = \left(\frac{h^2}{\mu} - \vec{r} \cdot \vec{e}\right)^2 = \frac{h^4}{\mu^2} + (\vec{r} \cdot \vec{e})(\vec{e} \cdot \vec{r}) - \frac{2h^2}{\mu}(\vec{r} \cdot \vec{e})$$

- We can write this as a quadratic form (equation for ellipse):

$$\vec{r} \cdot \vec{r} - (\vec{r} \cdot \vec{e})(\vec{e} \cdot \vec{r}) + 2\frac{h^2}{\mu}(\vec{e} \cdot \vec{r}) = \frac{h^4}{\mu^2}$$

$$\Rightarrow \tilde{r}^T \underbrace{(I - \tilde{e}\tilde{e}^T)}_{A} \tilde{r} + \underbrace{\left(2\frac{h^2}{\mu}\tilde{e}^T\right)}_{b} \tilde{r} = \underbrace{\frac{h^4}{\mu^2}}_{c}$$

$$\Rightarrow \tilde{r}^T A r + b^T r = c$$

$A = I - ee^T$ is symmetric and PSD as long as $\|e\| < 1 \Rightarrow$ defines an ellipse

Orbital Elements:

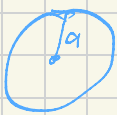
* Orbital elements are just a way of writing down the shape + orientation of an elliptical orbit.

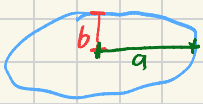
* There are many other ways with different pros/cons

* Assume a single massive body with a perfectly spherical gravity field.

Semimajor Axis (a):

- Units of length
- Defines size of orbit
- For a circular orbit $a = r$
- For an ellipse:





(b = "semiminor axis")

- For a parabola $a \longrightarrow \infty$

- For a hyperbola:



Eccentricity (e):

- Unitless
- Measures "deviation from circular"

$$e = \sqrt{1 - \frac{b^2}{a^2}}$$

$e = 0 \Rightarrow$ circular orbit
$0 < e < 1 \Rightarrow$ ellipse

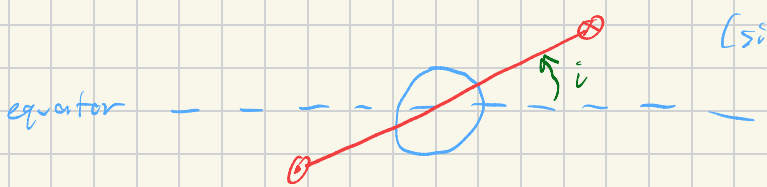
$\lim_{a \to \infty} e = 1$ for parabola

- Hyperbolic case!

$$e = \sqrt{1 + \frac{b^2}{a^2}} > 1$$

Inclination (i):

- Units of radians
- Angle between orbital plane and "reference plane" (e.g. equatorial plane)

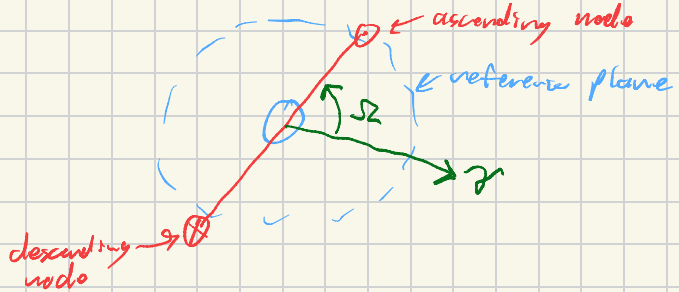


Longitude of the Ascending Node (RAAN) (Ω):

- Units of radians
- Rotation of the orbital plane about $\hat{z}$ axis (normal vector to the reference plane - think north pole)

- Measured relative to some reference direction. Convention is "first point of Aries" / "vernal equinox". Has symbol γ for "Ram's Head"

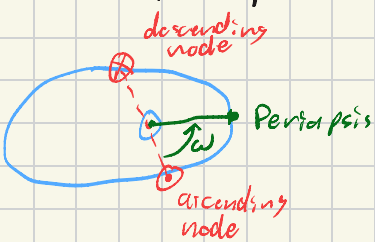
- Measure in right-hand direction from γ
- Top-down view with $i = 90^\circ$



Argument of Perigee / Periapsis (ω)

- Units of radians
- Rotates orbit about $\vec{h}$ vector (orbit normal)
- Measured from the ascending node to periapsis in direction of motion

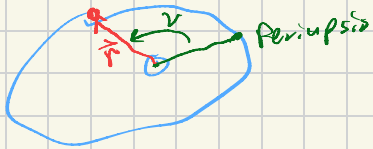
- Looking down at orbital plane:



True Anomaly (ν):

- Units of radians
- Gives the position of the spacecraft along the orbit at a given time.

- Measured from perigee to spacecraft position in the direction of motion


$\vec{v}$
$\vec{r}$
periapsis